

Prabin Thapa

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OBJECTIVE

Motivated and research-driven Electronics, Communication and Information Engineering graduate with a focus on applied AI for medical image analysis and intervention planning. Passionate about developing and applying AI-driven methods for medical image interpretation, diagnosis, and image reconstruction. Experienced in medical image processing, radiomics, and predictive modeling with a strong interest in integrating GenAI and computational analysis in biomedical data. Eager to contribute to innovative research that bridges computer science with biomedical applications to advance healthcare outcomes.

EDUCATION

Bachelor of Engineering (B.E) - Electronics, Communication and Information Engineering 2022 – 2026
Institute of Engineering, Tribhuvan University
Kathmandu, Nepal
Grade: 72.65%

SKILLS

IELTS Academic: Overall 7.0 [Listening: 8.0, Reading: 7.5, Writing: 6.5, Speaking: 6.5]

Languages: Python, C++

Libraries: PyTorch, MONAI, TensorFlow, NumPy, Scikit-learn, Pandas, OpenCV, Seaborn

Machine learning: Feature Engineering, Optimization, Modeling, EDA, Contrastive Learning, Medical Image Processing, Deep Learning

Tools: Git, VS Code, Jupyter Notebook, Conda, ITK-SNAP, 3D Slicer, MATLAB, Excel

Soft Skills: Problem-Solving, Communication, Mass Presentation, Leadership, Team Collaboration, Creativity

RESEARCH EXPERIENCE

Research Intern Mar 2025 - Mar 2026
NepAI Applied Mathematics and Informatics Institute for Research (NAAMII)
Supervisor: Dr. Taman Upadhaya (Project Scientist, Cedars-Sinai)

- Conducted medical image analysis involving CT image processing, 3D annotation, tumor segmentation, localization, and medical image synthesis.
- Extracted and analyzed radiomic and dosiomic features, with robustness assessment using image perturbation, intraclass correlation coefficient (ICC), and Bland–Altman analysis.
- Developed and optimized machine-learning and deep-learning pipelines for predictive modeling using clinical and medical imaging data.
- Performed statistical analysis, feature engineering, and preprocessing of clinical and medical imaging datasets for downstream modeling.

Undergraduate Research Student Oct 2025 - Jun 2026
Pashchimanchal Campus, Tribhuvan University

- Research on Medical Diagnosis using AI in collaboration with 4 different hospitals in Nepal
- Data-collection and surveys with patients at hospital (600 patients)
- Collaborate and coordinate with faculty, doctors, health workers and medical staff for data validation
- Conference Presentation at Gandaki University International Conference-January 3-5, 2026

TEACHING EXPERIENCE

Teaching Assistant – 6th Annual Nepal AI School (ANAI 2025)

Dec 2025 – Jan 2026

NepAI Applied Mathematics and Informatics Institute for Research (NAAMII)

- Served as a Teaching Assistant during hands-on lab sessions of the 10-day Nepal AI School.
- Supported participants in practical sessions covering computer vision, graph neural networks, transformers, and AI for medical applications.

PUBLICATIONS & MANUSCRIPTS

- **Thapa, P.**, Acharya, K., Baral, S., & Tiwari, H. N. *Personalised Risk Reduction and Toxicity Control in Treatment of Non-small Cell Lung Cancer. Manuscript in preparation.*

PROJECTS

Personalised Risk Reduction and Toxicity Control in Treatment of Non-Small Cell Lung Cancer (PRICISE)

- AI-based prediction of radiation-induced esophagitis and pneumonitis using CT images, radiomic, dosiomic, and clinical features
- Data preprocessing and feature engineering on a multicentre dataset of 451 NSCLC patients
- Implemented ML (MLP, SVM, Random Forest, LightGBM, LASSO) and 3D CNN models (ResNet, DenseNet, EfficientNet) with nested cross-validation and Optuna optimisation.
- Logistic Regression with Optuna achieved the best ROC-AUC of 0.662, slightly outperforming the best deep learning model (ResNet101, ROC-AUC 0.6561).

Non-Invasive Detection and Stage Classification of Anemia in Pregnant Women in Nepal Using Image of Conjunctiva of the Eye.

- Data collection with coordination with 4 different hospitals of Nepal from 437 different patients
- Data labelling and Image processing
- RoI extraction i.e Conjunctiva of the eye using U-Net architecture
- Stage classification of Anemia using extracted conjunctiva of the eye with an accuracy of 81.30%

MRI Reconstruction

- Implemented conventional reconstruction algorithms like STCR, GRAPPA on the reconstruction of Myocardial images from undersampled radial data
- Trained self-supervised model for brain image reconstruction with val loss:0.0000696, val psnr:32.01446, val ssim:0.91213
- Self guided DIP models for image reconstruction

Auto-segmentation of bone anatomy in CT images

- Studied different medical image segmentation algorithms
- Trained model for efficient segmentation of anatomy
- Optimised model for better Dice scores (achieved 0.983)
- Hausdorff95 Distance among real and segmented groundtruth in range of 1mm-2mm

Synthetic Medical Imaging using GAN.

- Deployed GAN model for medical image synthesis
- Fine tuned the model for realistic and sensitive image generation
- Diffusion model for label inpainting to generate segmentation labels

Variants Generation for image patterns.

- Transformer network for image generation
- Fine tuned stable diffusion model for variant generation

AWARDS AND ACHIEVEMENTS

- **University Grant Commission – Mini-Research Grant (2025)**
Awarded for the project 'Anemia Detection in Pregnant Women in Nepal Using Deep Learning'
- **OpenStreetMap(OSM) Hackfest, Runner Up (2023)**
Awarded for the Best **Tours and Travels Management WebApp** with AI integrated recommendation system.